

SEQUOIA CAPITAL

Generative AI's Act Two

From Technology Hammer to Real Value

By Sonya Huang, Pat Grady and GPT-4 | September 2023

An analysis of how generative AI is transitioning from novelty demos to end-to-end solutions that solve real human problems.

Six Decades of Technological Progress Converged to Enable Generative AI

6

DECADES

Moore's Law

Delivered the compute horsepower to process exaflops of data

Compute

4

DECADES

The Internet

Produced trillions of tokens of training data, accelerated by COVID

Data

2

DECADES

Mobile & Cloud

Put a supercomputer in every human's palm

Distribution

ChatGPT was the spark that ignited a Cambrian explosion of innovation

GenAI Startups Surpassed \$1B Revenue Faster Than SaaS Ever Did

>\$1B

in startup revenue

Months, not years — outpacing SaaS

ChatGPT

Fastest-growing application in history

Strong PMF among students and developers

Midjourney

Hundreds of millions in revenue

Achieved with a team of just 11 people

Character.ai

AI entertainment & companionship

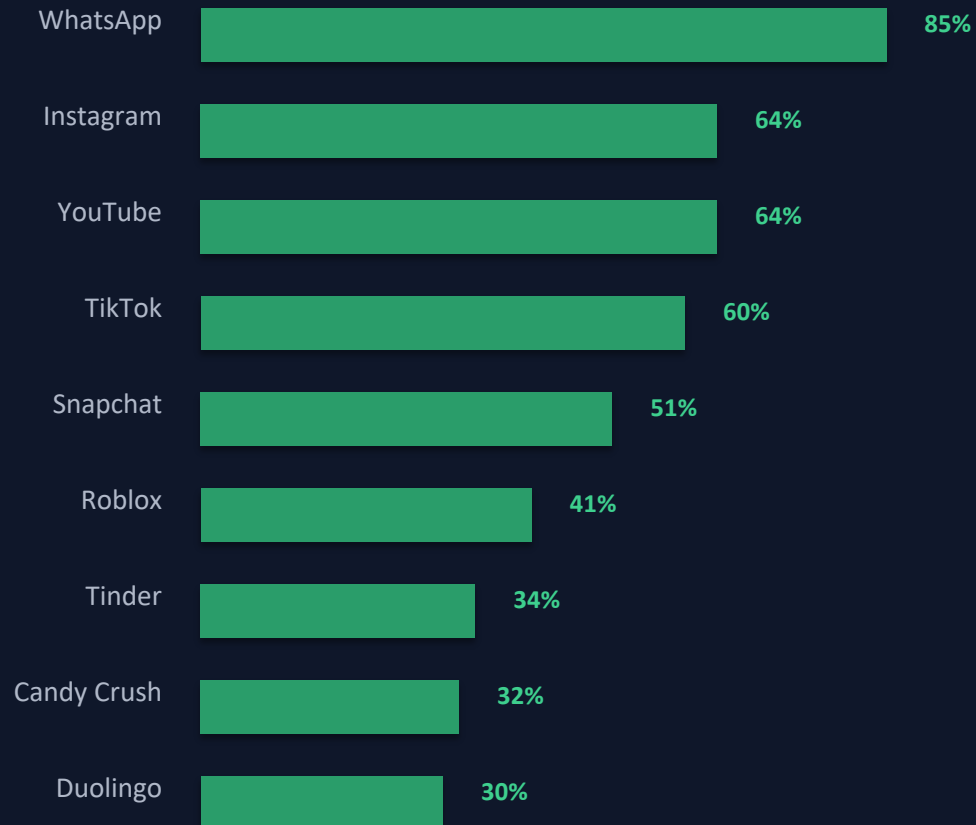
Users spend an average of 2 hours per session

Despite Traction, Cracks Appeared

- Every company became an "AI copilot" — inboxes filled with undifferentiated pitches
- Artists challenged machine-generated IP; Washington debated regulation
- Products fell short of expectations with terrible user retention
- The \$100M pre-product seed round returned amid hype
- Many AI companies still lack product-market fit or sustainable competitive advantage

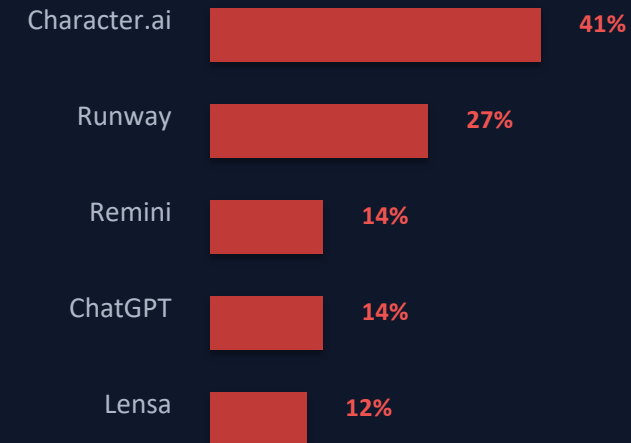
AI-First Apps Show 14% Median Retention vs. 51% for Incumbents

INCUMBENTS



Median: 51%

AI-FIRST COMPANIES



Median: 14%

3.6x retention disadvantage — AI-first apps must evolve beyond thin model wrappers to close this gap

YouTube Leads Incumbents at 85% Monthly Active Usage

Monthly active usage rates set the engagement bar AI apps must clear

Monthly Active Usage



Median: 63% — The engagement bar AI apps must clear

Act 1 Was Technology-Out; Act 2 Is Customer-Back

ACT 1: TECHNOLOGY-OUT

- Foundation models as the entire solution
- Novelty demos and lightweight wrappers
- Cool technology looking for a problem
- Single-modality applications
- Poor retention (14% median for AI-first apps)
- Thin competitive moats
- "AI copilot" for everything — undifferentiated

Result: Products fell short of expectations
with terrible user retention

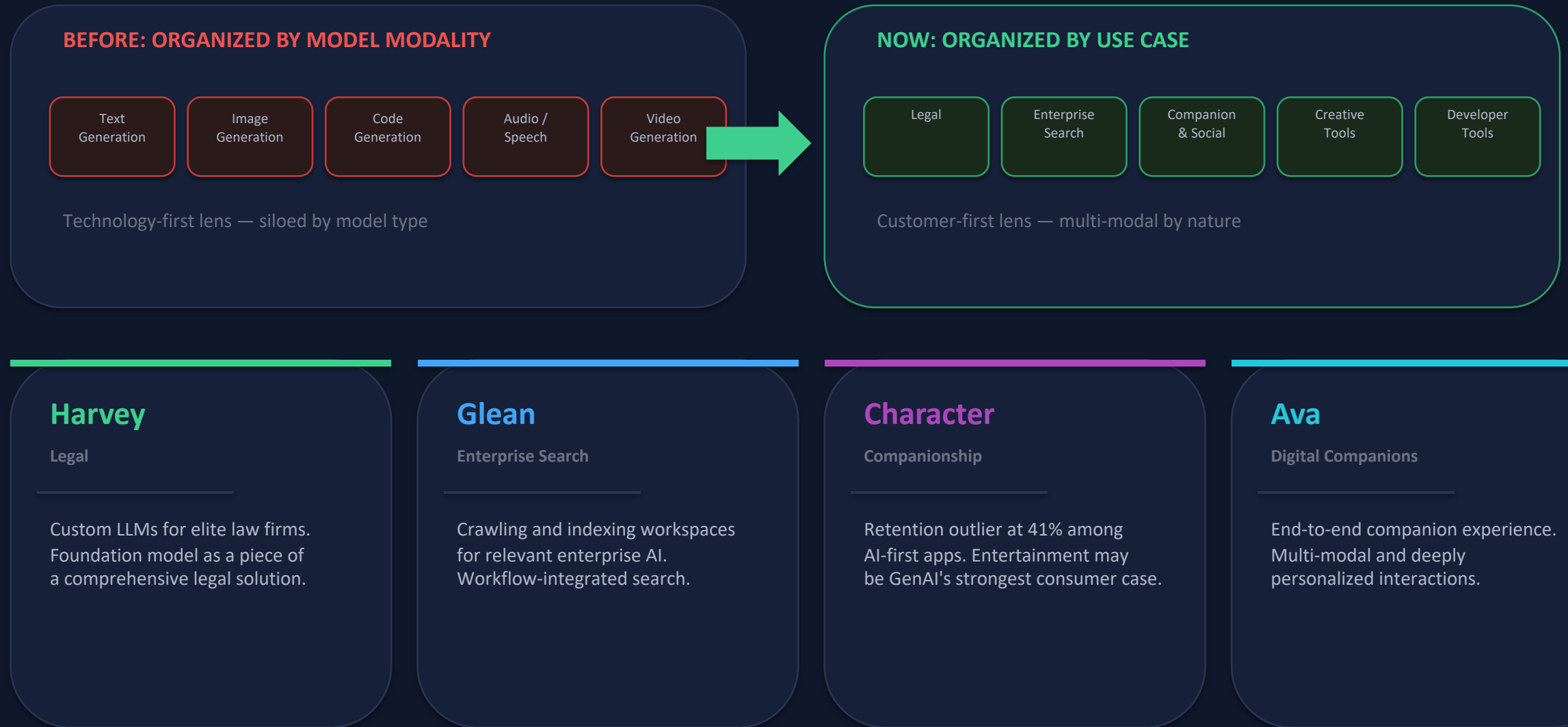
ACT 2: CUSTOMER-BACK

- Foundation models as a piece of a comprehensive solution
- New editing interfaces for stickier workflows
- Multi-modal in nature
- Solve problems end-to-end
- Workflow-integrated, not standalone
- Market organized by use case, not model modality
- Deep domain expertise as competitive moat

Act 2 Exemplars

Harvey (Legal) | Glean (Enterprise) | Character (Companions) | Ava
(Digital Companions)

Market Map Reorganized by Use Case, Not Model Modality



Key Takeaways: What Separates Winners in Act Two

01 Retention Is the Critical Challenge

AI-first apps at 14% median retention vs. 51% for incumbents — a 3.6x disadvantage. Closing this gap is existential for AI-first companies.

02 Foundation Models Are a Component, Not the Product

Act 2 winners use foundation models as a piece of a comprehensive solution, not the entire product. Thin wrappers won't survive.

03 Multi-Modal, Workflow-Integrated Solutions Will Win

New editing interfaces, end-to-end problem solving, and multi-modal capabilities create stickier products with defensible moats.

04 Real Demand Exists Despite the Hype Cycle

The >\$1B revenue milestone proves genuine commercial traction. GenAI outpaced SaaS's early trajectory — the opportunity is real.

05 Solve End-to-End Human Problems

Companies like Har

Act 2 has begun. The question is not whether AI will matter — but who will build it right.